

Proofs about sets & Quantification

Robert Y. Lewis

CS 0220 2021

February 11, 2022

Overview

1 Proof by Cases (1.7)

2 Predicate Formulas (3.6)

Fact about groups of people

Any two people have either met or not.

Given a group of people G , if all pairs of people in G have met, we'll call it a *club*. If no two people in G have met, we'll call them *strangers*.

Theorem. Every collection of 6 people includes a club of 3 people or a group of 3 strangers.

Does that seem true? Try some examples on the board.

Proof (Part 1)

The proof is by case analysis. Let x denote one of the six people. Let $R = G - \{x\}$ be the rest. There are two cases:

- 1 Among R , at least 3 have met x .
- 2 Among R , at least 3 have *not* met x .

At least one of these cases must hold. Since $|R|$ is odd, either more than half in R know x or less than half in R know x (and therefore more than half do not know x).

Case 1: At least 3 have met x . Let $J \subseteq R$ be those individuals. Two subcases:

- 1.1 No pair in J have met each other. So, J is a group of at least 3 strangers and the theorem holds in this subcase.
- 1.2 Some pair in J have met each other. That pair and x are a club of 3 people and the theorem holds in this subcase, too.

That covers Case 1!

Proof (Part 2)

Case 2: At least 3 have not met x . Let $J \subseteq R$ be those individuals. Two subcases:

- 2.1 Every pair in J have met each other. So, R is a club of at least size 3 and the theorem holds in this subcase.
- 2.2 Some pair in J haven't met each other. That pair and x are a group of strangers of 3 people and the theorem holds in this subcase, too.

That covers Case 2! It's kind of the inverse-video version of Case 1.

Since we showed that only these two cases can occur and the theorem holds in both, the theorem *always* holds.

Quantifiers, Revisited

Always True (universal quantification)

$\forall x \in \mathbb{R}, x^2 + 1 \geq 0.$

- For all $x \in D$, $P(x)$ is true.
- $P(x)$ is true for every x in the set D .

Sometimes True (existential quantification)

$\exists x \in \mathbb{Z}, x$ is even and x is prime.

- There is an $x \in D$ such that $P(x)$ is true.
- $P(x)$ is true for some x in the set D .
- $P(x)$ is true for at least one $x \in D$.

Mixing quantifiers

Theorem (sparse squares): There's a perfect square arbitrarily far from its closest perfect square.

Clear? Maybe a tad vague. True? How say in math?

$\forall d \in \mathbb{N}, \exists i \in \mathbb{N}, \forall j \in \mathbb{N},$
 $(i \text{ is a perfect square}) \wedge (|i - j| \leq d \rightarrow \neg(j \text{ is a perfect square})).$

The expressions nest inside each other. The order matters.

You can think of it like a little game. I'm claiming that you can pick any d you want. I'll then pick an i that's a perfect square AND no matter what j you pick that is within d values of i , j won't be a perfect square.

So, what's my winning strategy?

Any ambiguity is too many

“If you can identify any bird, you’ve got a talent.”

- 1 If $\exists b$, you can identify b , then you’ve got a talent.
- 2 If $\forall b$, you can identify b , then you’ve got a talent.

“...statistics show that, in the UK, a person brews a cup of tea every second. That person’s name is Nigel.”

- 1 $\forall t, \exists p, p$ brews a cup of tea at second t
- 2 $\exists p, \forall t, p$ brews a cup of tea at second t

More examples

“The whole article is not available.”

- not $\forall$ article part x , x is available
- $\forall$ article part x , x is not available

Data privacy: consider an “anonymous” survey with a number of questions. How to pool the responses?

- For each response r , for each answer a on r , there is a person p who gave answer a .
- For each response r , there is a person p such that for each answer a on r , p gave answer a .

DeMorgan returns: Negating quantifiers

These two statements are equivalent:

- Not everyone likes coffee.
- There's someone who doesn't like coffee.

$\neg \forall x, P(x)$ is equivalent to $\exists x, \neg P(x)$.

Assertion about predicates

The formula $\exists x, \forall y, P(x, y)$ implies the formula $\forall y, \exists x, P(x, y)$.

If $\exists x, \forall y, P(x, y)$, there must be some specific x^* such that $\forall y, P(x^*, y)$. As a result, $\forall y, \exists x, P(x, y)$ because we can always choose x^* to be the selected x .

On the other hand,

$\forall y, \exists x, P(x, y)$ does not imply $\exists x, \forall y, P(x, y)$.

Let's come up with a *counterexample*: a property P where the first formula is true and the second is false.

We can describe P by a table saying for which values of x and y $P(x, y)$ is true:

	y_1	y_2	y_3		y_1	y_2	y_3
x_1	T	F	F	x_1	T	F	F
x_2	T	T	T	x_2	T	F	T
x_3	F	T	F	x_3	F	T	F